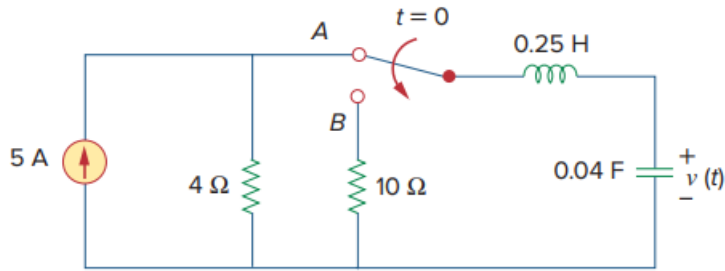


Problem

In the circuit of Fig. 8.71, the switch instantaneously moves from position *A* to *B* at $t = 0$. Find $v(t)$ for all $t \geq 0$.

Figure 8.71:



Step-by-step solution

Step 1 of 8

Refer to circuit diagram in Figure 8.71 in the textbook.

At $t = 0^-$ the switch is connected to position 'A'. The inductor acts like a short circuit, while the capacitor acts like an open circuit. The equivalent circuit is shown in Figure 1.

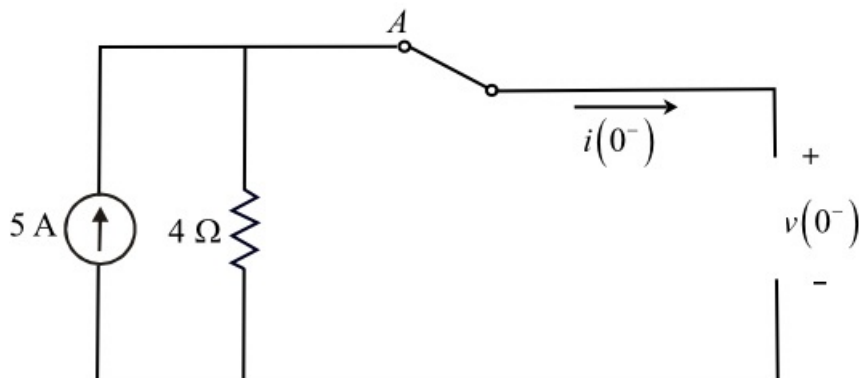


Figure 2

Step 2 of 8

As it is open circuited current is zero.

Hence, the initial value of current is $i(0) = 0 \text{ A}$.

Calculate the initial voltage across the capacitor using Ohm's law.

$$\begin{aligned} v(0^-) &= (4)(5) \\ &= 20 \text{ V} \end{aligned}$$

Since voltage across the capacitor cannot change instantaneously.

Therefore,

$$\begin{aligned} v(0) &= v(0^-) \\ &= 20 \text{ V} \end{aligned}$$

Step 3 of 8

At $t = 0^+$ the switch is connected to position 'B'. Draw the modified circuit diagram.

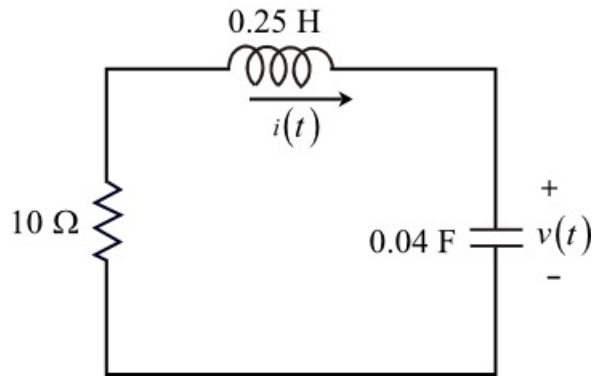


Figure 3

Step 4 of 8

Calculate the neper frequency.

$$\begin{aligned}\alpha &= \frac{R}{2L} \\ &= \frac{10}{(2)(0.25)} \\ &= 20\end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{0.25 \times 0.04}} \\ &= 10\end{aligned}$$

Step 5 of 8

Calculate the roots of characteristic equation.

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -20 \pm \sqrt{20^2 - 10^2} \\ &= -20 \pm 17.32 \\ &= -37.32, -2.68\end{aligned}$$

Since $\alpha > \omega_0$ the response is over damped

Therefore, the response is

$$\begin{aligned}v(t) &= A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ V} \\ v(t) &= A_1 e^{-2.68t} + A_2 e^{-37.32t} \text{ V} \dots\dots (1)\end{aligned}$$

Substitute 0 for t in equation (1).

$$\begin{aligned}v(0) &= A_1 e^{-2.68(0)} + A_2 e^{-37.32(0)} \text{ V} \\ v(0) &= A_1 + A_2 \\ A_1 + A_2 &= 20 \\ A_1 &= 20 - A_2 \dots\dots (2)\end{aligned}$$

Step 6 of 8

Write the expression for voltage across the capacitor.

$$v(t) = \frac{1}{C} \int i(t) dt$$

Differentiate the equation with respect to 't'.

$$\frac{dv(t)}{dt} = \frac{1}{C} i(t)$$

Substitute 0 for i in the equation.

$$\begin{aligned} \frac{dv(0)}{dt} &= \frac{1}{C} i(0) \\ &= \frac{1}{(0.25)} (0) \\ &= 0 \text{ V/s} \end{aligned}$$

Step 7 of 8

Recall equation (1).

$$v(t) = A_1 e^{-2.68t} + A_2 e^{-37.32t} \text{ V}$$

Differentiate the equation with respect to 't'.

$$\begin{aligned} \frac{dv(t)}{dt} &= -2.68 A_1 e^{-2.68t} - 37.32 A_2 e^{-37.32t} \\ \left. \frac{dv(t)}{dt} \right|_{t=0} &= -2.68 A_1 - 37.32 A_2 \\ -2.68 A_1 - 37.32 A_2 &= 0 \end{aligned}$$

Substitute $20 - A_2$ for A_1 in the equation.

$$\begin{aligned} -2.68(20 - A_2) - 37.32 A_2 &= 0 \\ -53.6 + 2.68 A_2 - 37.32 A_2 &= 0 \\ -34.64 A_2 &= 53.6 \\ A_2 &= -1.547 \end{aligned}$$

Step 8 of 8

Recall equation (2).

$$A_1 = 20 - A_2$$

Substitute 12 for A_2 in the equation.

$$\begin{aligned} A_1 &= 20 - (-1.547) \\ &= 21.547 \end{aligned}$$

Recall equation (1).

$$v(t) = A_1 e^{-2.68t} + A_2 e^{-37.32t} \text{ V}$$

Substitute 21.547 for A_1 , 12 for A_2 in the equation.

$$v(t) = 21.547 e^{-2.68t} - 1.547 e^{-37.32t} \text{ V}$$

Therefore, the voltage $v(t)$ for $t \geq 0$ is $\boxed{21.547 e^{-2.68t} - 1.547 e^{-37.32t} \text{ V}}$.